TITAN-X

Executive Business Plan · 2026–2030

Stratospheric persistent aerial platform for venture capitalists and business owners — regional infrastructure between aircraft and satellites.

Firefighting configuration — retardant and water delivery. California is the planning early concentration.

Planning estimates for discussion with qualified parties. Not guarantees of future performance. Not an offer to sell securities.

1. Executive summary

TITAN-X is a free-flight stratospheric aerostat with a **20,000 lb** modular payload, indefinite endurance on helium buoyancy and solar power, and mission reconfiguration by swapping gondolas and underbelly modules.

Investment thesis: fund pathfinder → first full-scale hull → California firefighting + metro nodes → planning **12-unit** fleet by 2030. Pre-positioning converts response from days to hours; civil and commercial utilization underwrites year-round availability; dual-use configs share one cost base.

$13M	**$156M**	**$2.4M**	**$200K**	**17.8 : 1**	**20,000 lb**
Unit CAPEX (fleet volume)	12-unit fleet CAPEX	Annual OPEX / unit	Monthly break-even / unit	Planning 10-yr fleet ROI	Net modular payload

2. Platform at a glance

Parameter	Value	Notes
Size	480 ft × 200 ft	Swept manta planform
Net payload	20,000 lbs	≈ school-bus weight
Altitude range	500 – 130,000 ft	Tree-top to lower stratosphere
Endurance	Indefinite	Solar + buoyancy
Ops carbon (persistent)	Near-zero	No combustion for loiter
Architecture	Fully modular	Gondola / underbelly swap

- Rockoon loft to ~100,000 ft for propellant savings, then recover the platform
- Civil, commercial, and defense requirements share the same platform and cost base

Formation / persistent high-altitude operations — regional pre-positioning on one modular hull family.

3. Problem & solution

Wildfire and disaster response still depends on scarce surge aircraft. Satellites persist but cannot deliver mass. Cities need hours-scale airborne nodes for observation, temporary communications, and logistics.

TITAN-X fills the band between aircraft and satellites: indefinite loiter, school-bus-class modular payload, near-zero persistent ops carbon, and pre-positioned regional coverage instead of surge-only response.

4. Regional deployment

Major cities & high-value regions

One TITAN-X (or a pair) per region, loaded with modular response packages, on station in hours — disaster response, communications restoration, observation, logistics.

California firefighting concentration

- Fire-behavior, thermal, and smoke-plume sensors
- Retardant or water delivery modules
- Incident command and communications packages
- Decision-support for incident commanders

Modules shift to other missions in lower-risk periods to preserve utilization.

Smaller complementary platforms

Secondary observation, relay, weather, and lower-cost coverage under full-size regional nodes.

5. Customers & missions

Segment	Use
State / regional wildfire authorities	CA firefighting concentration
Metro & high-value regions	Persistent disaster & comms nodes
Critical infrastructure	Monitoring and logistics support
Maritime / shipping	Domain awareness and ship support
Public safety	Airborne threat awareness (low-cost drones)
Science / weather	High-altitude observation
Rockoon / launch customers	Stratospheric loft + recover
Dual-use / defense (selected)	ISR, relay, forward logistics, responsive access

6. Unit economics

Item	Planning value
Unit CAPEX (fleet volume)	$13.0M
12-unit fleet CAPEX	$156M
Annual OPEX per unit	$2.4M
Break-even utilization	~$200K / month
Mature utilization target (base)	$8–12M / unit / year
Contribution at $10M rev / $2.4M opex	~$7.6M / unit / year

Illustrative mature revenue mix

Stream	Share
Regional standby / availability	35–45%
Wildfire / disaster mission fees	20–30%
Infrastructure / maritime / public safety	15–25%
Rockoon / science / specialty	5–15%
Dual-use / defense options	0–20%

7. Financial plan 2026–2030

USD millions. Anchored to consolidated summary CAPEX / OPEX / break-even. The planning 17.8 : 1 ten-year fleet ROI is a long-horizon target on the stood-up fleet — not a 2030 cash guarantee.

Fleet build (base)

	2026	2027	2028	2029	2030
Full-size units EOY	0*	1	3	6	12
CA firefighting-primary	—	1	2	3	4
Metro / regional nodes	—	0	1	3	8

*Pathfinder year — endurance, station-keeping, modular swap validation.

Capital expenditure (base, $M)

	2026	2027	2028	2029	2030	Total
Pathfinder + engineering	18	6	4	3	2	33
Hull production (learning curve)	—	18	30	42	78	168
Ground / ops infrastructure	4	6	8	10	12	40
Total CAPEX	22	30	42	55	92	241

Operating model (base, $M)

	2026	2027	2028	2029	2030

Avg units in service	0	0.5	2.0	4.5	9.0
Revenue	0.5	4	18	42	90
Direct OPEX	1.5	3.5	7	13	24
Program / G&A	4	5	7	9	11
EBITDA (planning)	(5.0)	(4.5)	4	20	55

2030 check: ~$90M revenue on ~9 average units ≈ **$10M per unit-year**, above the $2.4M OPEX break-even.

Scenario fan — 2030

Case	Units	Revenue	EBITDA
Conservative	6	$45M	$18M
Base	12	$90M	$55M
Upside	12+	$130M	$85M

8. Use of funds & milestones

Use	Share
Pathfinder + first full-scale hull	35–45%
CA firefighting + first metro modules	20–30%
Production tooling / learning-curve	15–20%
Ops bases, helium, insurance, crew	10–15%
Corporate / BD / compliance	5–10%

Gate heavy fleet CAPEX on endurance demo, modular swap demo, and first paid availability or mission contract.

9. Risks

- Airworthiness / airspace — pathfinder before fleet claims
- Early unit cost above $13M volume target — learning-curve modeled explicitly
- Public contracting cycles — civil commercial utilization underwrites standby
- 17.8 : 1 ROI is a planning ten-year fleet figure, not a 2026–2030 guarantee

10. The ask

We are engaging venture capitalists and business owners to fund or co-develop:

- Pathfinder validation and first full-scale hull
- California firefighting concentration as the first revenue-bearing mission stack
- Metro pre-positioning nodes that turn TITAN-X into infrastructure

Bottom line. Indefinite endurance. 20,000 lb modular payload. Near-zero persistent ops carbon. Pre-positioned regional coverage with a California firefighting emphasis. Dual-use economics on one hull family. **2026–2030 builds the twelve-unit base; utilization after that is where the summary ROI lives.**